# Deepen or Diversify?

## Estimating Firm Capability Reconfiguration under Trade Shocks

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**The project in one paragraph.** When a trade shock degrades the value of a firm's existing activities, does the firm *retrench* toward a coherent capability core, or *reinvent* itself by exploring new capability bundles? I measure the firm's export basket as a position in a firm–product *capability network*, split diversification into an *in-block* margin $d_{\text{in}}$ (deepening the core) and an *out-of-block* margin $d_{\text{out}}$ (exploring beyond it), and ask how a predetermined, continuous tariff shock moves the two. The static capability map is already built on Chinese customs microdata (§2); the causal panel is the next step (§3). Sections 1–3 lay out the design and its problems; the open questions for discussion are collected in §4.

## Contents



# 1 The question and why it is open

The empirical trade literature has measured, in great detail, how tariff shocks move firm *performance*: productivity, markups, employment, survival (Pavcnik 2002; Topalova 2010; Amiti–Konings 2007; Le & Tomasi 2023). It is almost silent on what those shocks do to the *composition* of the firm's activity — which products are kept, dropped, or newly entered, and whether the surviving basket is a *deepening* of a related core or a *scattering* into new domains. This is the object I want to make causal.

The reason the sign is genuinely open is theoretical. March's (1991) exploitation–exploration tension says a firm whose core is threatened can either intensify what it knows (retrench) or search outward (explore), and his own result is that organisations systematically *over-exploit*. Several literatures sharpen this into a directional prediction:

- **Retrenchment (H1, the modal prior).** Sunk-cost and re-entry-cost hysteresis (Crozet et al. 2021), variance-minimisation through within-cluster deepening (Bottazzi–Secchi 2006;

Vannoorenberghe 2016; Crozet et al. 2025), and the competition-to-coherence evidence of Valvano–Vannoni (2003) and the product-vector consolidation of Fontagné et al. (2018) all point to $d_{\text{in}}\uparrow$, $d_{\text{out}}\downarrow$, Share_core$\uparrow$.

- **Reinvention (H1′, the frontier-firm tail).** Penrose (1959), Teece (1982), the moderate-shock search models of Matsusaka (2001) and Bernardo–Chowdhry (2002), and escape-competition (Aghion et al. 2005; Akcigit–Melitz 2022) leave open $d_{\text{out}}\uparrow$ for firms with slack and a defensible core.

The economic-complexity toolkit (Hidalgo–Hausmann 2009; Tacchella et al. 2012; Pugliese et al. 2019; Bruno et al. 2023; De Stefano et al. 2025) gives me the *measurement instrument* for capability structure — a block-modular firm–product partition — but it has been used almost exclusively descriptively. The strategic-dependency literature (Farrell–Newman 2019; Arjona et al. 2023; Korniyenko et al. 2017) identifies *which* products carry geopolitical leverage but has no firm-level causal test. The contribution I am aiming for sits in the empty intersection: *a causal estimate of the direction of capability adjustment to a predetermined trade shock, with the strategic-dependence content of the product as the key moderator.*

## 2 Preliminary evidence: the capability map of Chinese exporters

Before any causal claim, the measurement layer has to be real and economically legible. I build it on Chinese firm×product×destination×year customs exports. For the 2008–2015 window the firm–product–year panel has 11.9M rows over 372,030 firms and 4,657 HS6 products. To work with an established-exporter core I keep firms that export in *every* year of the window: 71,017 firms — only 19% of the count, but 62% of export value. The export basket is binarised with the firm-level Balassa index ($\text{RCA}_{fp} > 1$), giving a bipartite incidence with ∼1.1M links. Both firm diversification and product ubiquity are heavily right-skewed (Fig. 5 in the appendix), the usual complexity signature.

**How the capability blocks are found (the algorithm, in plain terms).** The firm–product matrix is a *bipartite* graph: firms on one side, products on the other, an edge whenever a firm has RCA $> 1$ in a product. A "capability block" is a set of firms *and* products that link among themselves far more densely than chance would predict. "Than chance" is made precise by a null model. Bipartite modularity $Q_B$ (Barber 2007) scores a partition by the within-block edges *in excess* of those expected if every firm and every product kept its own number of links but rewired at random — the expected edge between firm $f$ and product $p$ is $k_f d_p/m$ (degree×degree over total links). BRIM is the heuristic that maximises $Q_B$. The *bipartite* null is the whole point: off-the-shelf optimisers (Louvain, Leiden) use a *one-mode* null that "expects" firm–firm and product–product edges which cannot exist in a bipartite graph; that mis-specified baseline inflates apparent within-block density and *over-segments* (here 13 and 9 blocks against BRIM's 5). A resolution parameter $\gamma$ scales the null "charge" for grouping two nodes: $\gamma = 1$ is standard modularity, $\gamma > 1$ charges more and returns more, smaller blocks. I use BRIM at $\gamma = 1$ as the baseline and report $\gamma = 2$ below.

**Capability blocks are real, and they are not HS sectors.** Maximising Barber's (2007) bipartite modularity (BRIM) on the incidence matrix yields **five capability blocks** with a strong modular signal, $Q_B = 0.505$. The blocks have a clear economic identity (Table 1, Fig. 1): a large machinery and electrical *generalist* block (51% of firms, the assembly base), a paper/light-manufacturing block (23%), a near-pure *textiles* block (15%), a chemicals-plus-agri-food block (11%), and a tiny ultra-specialised diamond satellite. The data-driven partition is *not* a relabelling of HS sectors: against the HS-section partition it scores ARI = 0.29, NMI = 0.43, with blocks spanning on average thirteen HS sections. The block is a capability object, not an industry code — which is exactly what makes the $d_{\text{in}}/d_{\text{out}}$ split meaningful.

**Table 1:** The five capability blocks (2008–2015, HS6; 71,017 established exporters).

| Block | Firms | Dominant content |
|---|---|---|
| B0 | 36,321 (51%) | Machinery & electrical 36%, base metals 24%, instruments 7% — the large *generalist* block |
| B2 | 16,095 (23%) | Paper 16%, misc. manufacturing, packaging — light manufacturing |
| B1 | 10,870 (15%) | **Textiles 95%** — apparel, cotton, man-made fibres (pure) |
| B3 | 7,691 (11%) | **Chemicals 48% + agri-food ∼41%** (vegetables, foodstuffs, animals) |
| B4 | 40 (0%) | Precious 100% — diamonds (pure satellite) |

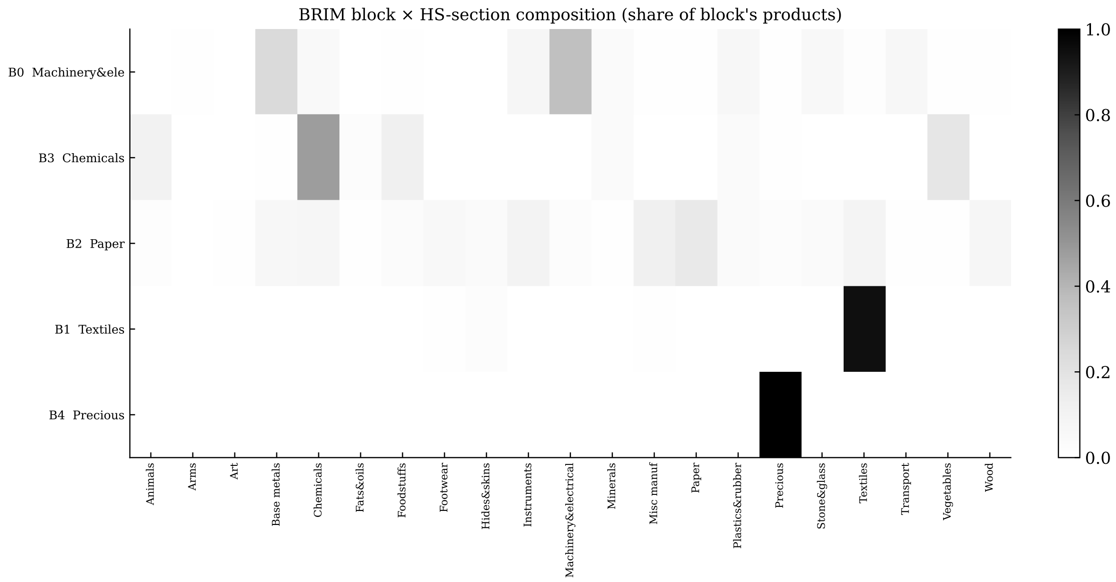


**Figure 1:** Capability block × HS-section composition. Textiles (B1) is near-pure; the generalist block (B0) spreads across machinery, metals and instruments. The off-diagonal mass is the evidence that blocks are not industry codes.

**Does the resolution matter? ($\gamma = 1$ vs $\gamma = 2$).** The block *count* is not an invariant — it depends on $\gamma$. To check that the substantive findings do not hinge on it, I re-ran the entire pipeline at the finer $\gamma = 2$, which penalises the null more and so returns more, smaller blocks (Table 2, Fig. 2). Some of the split is *economically meaningful*: agri-food, bundled with chemicals at $\gamma = 1$, separates into its own block, and dedicated footwear and transport blocks emerge. Some is *mechanical*: the pure textile block splits into two near-identical sub-blocks, while the machinery–metals generalist stays mixed at any resolution (it is where the bulk of firms sit). The point is that the *conclusions are invariant and only the levels rescale*: the structure is strongly modular and not a relabelling of HS sectors at both resolutions, and out-of-block exploration tracks destination reach rather than export value at both. What does move is the in/out accounting — finer blocks mean smaller in-block menus, so the out-of-block share mechanically rises (pure-in-block 54.5%→43.2%). The lesson I carry to the panel: *fix one partition choice and read $d_{\text{out}}$ within a resolution, never across.*

**Table 2:** The same pipeline at two resolutions. Levels rescale; conclusions do not.

| Quantity | $\gamma = 1$ (baseline) | $\gamma = 2$ |
|---|---|---|
| BRIM blocks | 5 | 10 |
| Bipartite modularity $Q_B$ | 0.505 | 0.544 |
| ARI vs HS sections | 0.29 | 0.32 |
| HS sections spanned per block (avg) | 13 | 9.4 |
| Pure in-block (Share_core = 1) | 54.5% | 43.2% |
| corr($d_{\text{out}}$, dest. reach) | 0.18 | 0.16 |
| corr($d_{\text{out}}$, export value) | 0.03 | 0.06 |

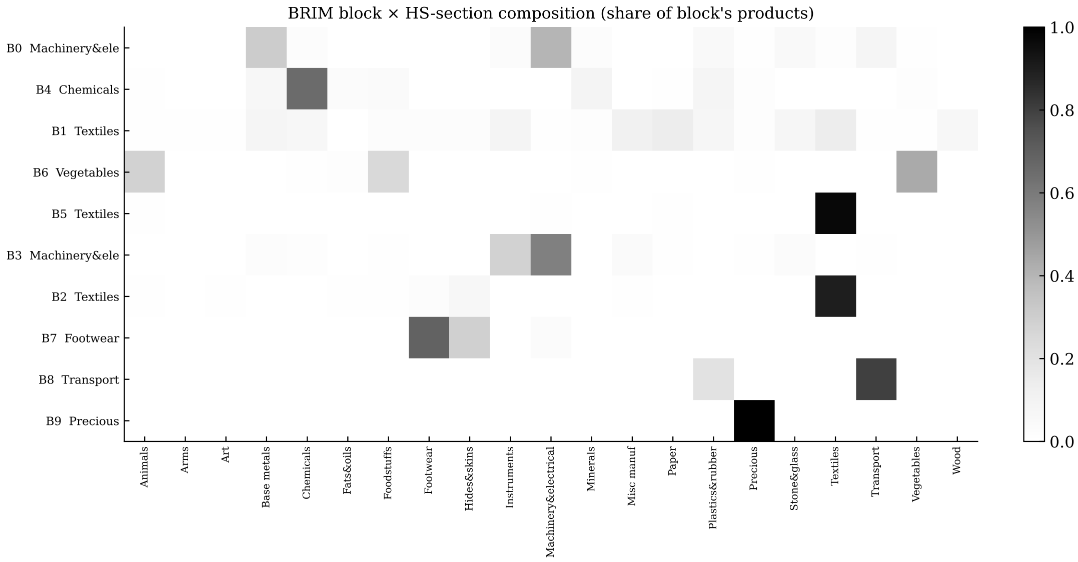


**Figure 2:** $\gamma = 2$: the ten finer blocks × HS sections (compare with the five-block $\gamma = 1$ map, Fig. 1). The newly isolated dark cells — a second textile block, a separated chemicals block, footwear, transport, precious — are the meaningful splits; the machinery generalist (and the light-manufacturing catch-all) stay spread, because they are genuinely entangled at any resolution.

**These exporters are exploitation-heavy.** Splitting diversification into $d_{\text{in}}$ (competitive products *inside* the firm's own block) and $d_{\text{out}}$ (those *outside* it), the picture is one of deep specialisation (Table 3, Fig. 3): mean $d_{\text{in}} = 13.0$ against mean $d_{\text{out}} = 2.5$, and the out-of-block margin is *exactly zero for 54.5% of firms.* The concentration index Share_core $= d_{\text{in}}/(d_{\text{in}} + d_{\text{out}})$ has a median of 1.0 and a large mass at the ceiling. In March's language, the established Chinese exporter grows along the $d_{\text{in}}$ axis, within its capability block. **That distributional fact is what most shapes the panel design**: $d_{\text{out}}$ is zero-inflated and floor/ceiling-bound, so the "reinvention" falsifier ($d_{\text{out}} \uparrow$) is better-powered than the "retrenchment" direction, and the outcome models have to respect the count/zero-mass structure (see §3).

**Table 3:** Diversification indicators ($N = 71{,}017$). $d_{\text{out}} = 0$ for the majority.

| | mean | sd | p10 | p25 | median | p75 | p90 |
|---|---|---|---|---|---|---|---|
| $d_{\text{in}}$ | 13.01 | 20.83 | 2 | 4 | 7 | 14 | 29 |
| $d_{\text{out}}$ | 2.53 | 11.38 | 0 | 0 | 0 | 2 | 5 |
| Share_core | 0.888 | 0.162 | 0.63 | 0.80 | 1.00 | 1.00 | 1.00 |

Pure in-block (Share_core = 1): 54.5%. Mixed: 45.5%. Pure out-of-block: 0%.

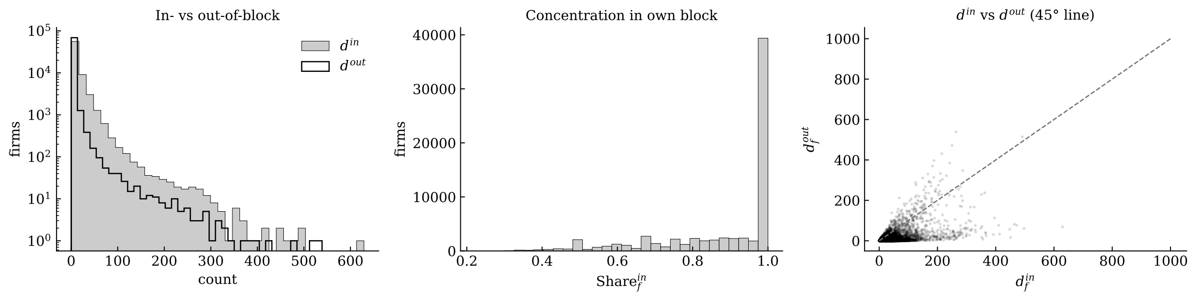


**Figure 3: Left:** $d_{\text{in}}$ vs $d_{\text{out}}$ (log $y$); $d_{\text{in}}$ dominates, $d_{\text{out}} = 0$ for the median firm. **Centre:** Share_core, with a mass at 1 (pure specialists). **Right:** $d_{\text{in}}$ vs $d_{\text{out}}$ with the 45° line — the cloud lies below it, firms grow within their block.

**A second pattern: out-of-block exploration is a *geographic* signature, not a size one.** Because the microdata carry the destination, I can give $d_{\text{out}}$ an economic identity. Across 71,017 firms, $d_{\text{out}}$ correlates with *destination reach* (number of countries served) at 0.18, but with *export value* at only 0.03 (Spearman). Reaching more markets tracks out-of-block exploration; sheer revenue does not (Fig. 4). Holding destination reach fixed, scaling up in value actually *lowers* the out-of-block share (it deepens the core): a clean monotone pattern at the baseline resolution, though the one finding that partially reverses under a finer partition, so I read it cautiously. The message that survives: **the "explorer" firm is the wide-reach exporter, not the big-revenue one**, and exploration is concentrated in particular blocks (the paper/light-manufacturing block explores three times as much as any other). That is what tells me the eventual design must track the product and the geographic margins *jointly*, and must guard against export intermediaries, whose unrelated baskets produce spuriously high $d_{\text{out}}$ and low portfolio coherence.

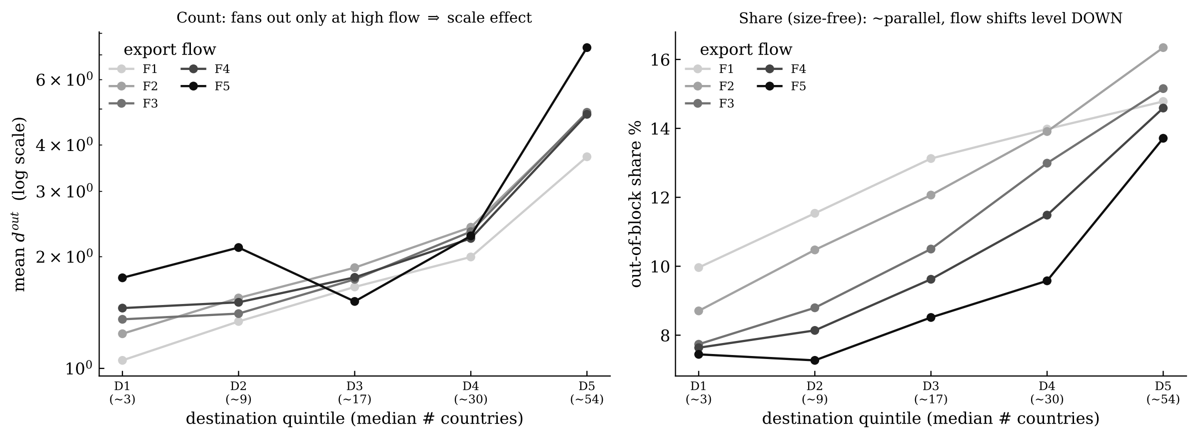


**Figure 4:** Out-of-block exploration by export-flow quintile (light = low to dark = high) × destination reach. **Left:** mean $d_{\text{out}}$ count (log) — rises with destinations. **Right:** size-free out-of-block share — again rises with destinations, with the high-flow lines sitting *below*. Breadth tilts the portfolio outward; volume pulls it inward.

# 3 From description to causation: the design and its problems

The static layer is descriptive. The causal question (does an adverse trade shock push the firm toward $d_{\text{in}}$ or $d_{\text{out}}$?) needs a firm-by-year panel of the outcomes and a credibly exogenous shock. Two of the three moving parts I think are tractable; the third is genuinely open. I lay out the problem and my candidate fixes here; the open questions they raise are collected in §4.

**The shock — I think this part is in hand.** I plan a continuous Bartik tariff dose

$$\text{TariffShock}_{it} = \sum_p \underbrace{s_{ip,\text{pre}}}_{\text{pre-shock shares}} \cdot \underbrace{\varepsilon_p}_{\text{Fontagné et al. 2022}} \cdot \underbrace{\Delta \ln(1+\tau_{pt})}_{\text{Teti GTD 2024}},$$

estimated through a distributed-lag event study (leads = parallel-trends test, lags = hysteresis) with Adão et al. (2019) shift-share-robust standard errors. The exposure shares are pre-shock and the elasticities $\varepsilon_p$ are external product parameters estimated on the world trade matrix, so they cannot be a function of the Chinese firms I study — this side of the design is predetermined by construction. I keep the dose *continuous* on purpose: discretising into treated/control throws away the exposure-intensity variation that identifies the dose–response (the cost is a stronger parallel-trends requirement, which I can probe but not fully test).

**The hard part — turning a static map into a panel.** Here is the crux, and the reason I want the algorithm understood (§2). The capability blocks are estimated on the *pooled* network. A panel needs the outcome to vary over time, but $d_{\text{in}}/d_{\text{out}}$ are defined *relative to the partition*, so the instant the partition itself moves, the outcome and its own measuring instrument move together. There are two routes, each with a cost:

- **(a) Re-estimate the partition every year.** Conceptually attractive (capabilities really do evolve), but it makes block membership co-move with the shock: a tariff that pushes a firm off a product can show up as the *product* switching blocks, so the treatment is absorbed *into* the measuring instrument instead of being read *against* it. Worse, the temporal-coupled multilayer machinery (Mucha et al. 2010) one would use to smooth the yearly slices *couples adjacent years*, so the block at $t$ depends on $t+1$: a look-ahead that disqualifies it as a clean right-hand-side panel object. (It is also computationally heavy: BRIM on each yearly slice at the full ∼460k-firm scale.)
- **(b) Estimate the partition once on pre-shock years and hold it fixed,** letting only the firm's RCA incidence vary. This keeps the measuring instrument exogenous to the shock, at the cost of a map that grows stale — classical measurement error, which attenuates *toward* zero. That is an honest bias, not a confound, but it is a real loss of power.

My current lean is (b) as the baseline, with a *pre-shock-evolving* map (estimated only on pre-shock slices) as robustness, and — the option I am most drawn to but least sure of — a binary, staggered design whose *not-yet-treated* firms could anchor a partition that evolves *contemporaneously yet stays exogenous*, in the spirit of a clean control group. The open choice is whether the cleaner estimand is worth a held-fixed (and therefore stale) map, or whether the capability geometry can be made to move *with* the data without moving *with the treatment.* Concretely it is a question about the *detection algorithm* itself: whether BRIM (or any community-detection method) can deliver a partition that updates with the data yet stays, by construction, blind to the shock. I take this back up in §4.

**What the preliminary data already dictate (whatever I decide above).** The static layer is not just motivation; it pins down choices. (i) $d_{\text{in}}$ is the cleanest outcome; $d_{\text{out}}$ is 54.5% zeros ⇒ a count/PPML model, not OLS. (ii) The ceiling at Share_core = 1 means the retrenchment direction is poorly powered and the $d_{\text{out}}\uparrow$ falsifier is the sharper test. (iii) Effects should vary by destination reach and by block, since that is where exploration lives. (iv) Coherence flags intermediaries (unrelated baskets, spurious $d_{\text{out}}$) and must be a control. (v) The balanced core is economically dominant but selected — external validity to the entry/exit tail is a stated caveat, and it interacts with the exit problem below.

# 4 Open issues, and the questions I would like your view on

I separate what I already see as the *soft spots* from the *questions* I would genuinely like to put to you. I have grouped the questions by mechanism, by empirical design, and by theory, but the most valuable feedback would be on which of these is the binding constraint on the project.

### Soft spots I am already worried about

1. **Exit selection.** The shock induces exit (Crozet et al. 2021); exited firms have no $d_{\text{in}}/d_{\text{out}}$, so survivors are selected. I plan an exit equation plus Lee/IPW bounds and read the headline as conservative — but the balanced-sample design already conditions on survival, which may understate the very margin I care about.
2. **Spillovers / SUTVA.** A tariff on one product reallocates demand to substitutes held by *other* firms in the same block; Adao et al. (2019) absorbs the correlation in the SE but not the causal interference.
3. **Zero-inflation and power.** With $d_{\text{out}} = 0$ for the majority, the retrenchment direction may be statistically hard to move; the design may end up better at *rejecting* reinvention than *confirming* retrenchment.

### (A) Questions on the mechanism

1. Is retrenchment to the core best read as *defensive pruning* (shedding the cheapest-to-cut peripheral products) or *active consolidation* (reinvesting in the core)? These have the same Share_core signature but opposite welfare readings. Is there a margin in the data — e.g. the complexity of the products *dropped* (Nomaler–Verspagen 2023) — that can separate them?
2. The destination finding suggests $d_{\text{out}}$ is partly a *geographic* object. Should the unit of analysis be the firm–product matrix (following the capability-network tradition) or a firm–product–destination tensor? Does promoting to the tensor buy identification, or just noise?
3. Is "exploration" through $d_{\text{out}}$ genuinely capability-building, or is a large share of it intermediary churn that coherence controls cannot fully remove?

### (B) Questions on the empirical design / how to innovate the analysis

1. **The held-fixed-vs-evolving map trade-off** (the open seam of §3). Beyond the two routes I sketched, is there a third I am missing — for instance a *partition-free* outcome (a continuous relatedness or coherence score of the basket) that would sidestep the timing problem altogether, at the cost of losing the crisp in/out split? And on the method itself: can a community-detection partition be made to evolve with the data yet stay blind to the shock, or is that the wrong way to frame it? Which would you reach for?
2. **Parallel trends for a continuous dose.** Beyond plotting lead coefficients, what is the most convincing way to defend "no selection into dose level" when the dose is a smooth shift-share and there is no clean control group?
3. Given the zero-inflation, is there a better-powered *single* outcome (a product-level retention hazard, in the spirit of Fontagné et al. 2018) than the firm-level Share_core?

### (C) A theoretical model — here I am more interested in what occurs to you

On the theory side I have instincts but no commitment, and this is where your reaction matters most — so these are deliberately open.

1. If you wanted to rationalise an in-block/out-of-block margin, *what backbone would you reach for*? My own instinct drifts toward a multi-product firm in the "core competence" tradition (Eckel–Neary 2010; Mayer–Melitz–Ottaviano 2014), where products sit on a ladder ranked by distance from the firm's core competence and tougher competition skews the mix *toward* the core — which is exactly my retrenchment prediction, with in-block ≈ near-core and out-of-block ≈ far-from-core. The honest gap is that their ladder is ordered by *cost/productivity*, whereas my capability blocks are ordered by *relatedness* (network co-occurrence): is "distance from core" the same object in the two senses, or do I need a genuine capability/relatedness model instead? That is the piece I would most like your reaction to.
2. Is there a natural way to make the *sign* endogenous — retrenchment for most firms, reinvention for a high-capability tail — as a comparative static in a single state variable (capability stock, or distance-to-core)? That is the form in which March's (1991) verbal tension would become testable, but I do not yet see the minimal model that gives it.
3. Does any of this *need* general equilibrium? A sector-wide tariff shock is an aggregate phenomenon, so its firm-level effects might be better captured inside a quantitative trade model — which would also let me speak to the *welfare* of retrenchment, not just its sign. Is that worth the modelling cost, or a distraction from a clean reduced-form paper?
4. If you did build it, which empirical moment would you discipline it on first — the dose–response slope, the $d_{\text{in}}/d_{\text{out}}$ heterogeneity by strategic dependence, or the destination-reach gradient?

**What I would most like to leave the meeting with.** A judgement on the binding constraint: is this project limited by *identification* (the map-timing choice and the continuous dose), by *measurement* (intermediaries and the balanced-sample selection), or by the absence of a *theoretical* object that makes the $d_{\text{in}}/d_{\text{out}}$ split a structural quantity? My current bet is that the empirics can carry a first paper on their own, with the model as a second-stage companion — but I would like to pressure-test that ordering with you.

# A Supporting figure: heavy-tailed structure

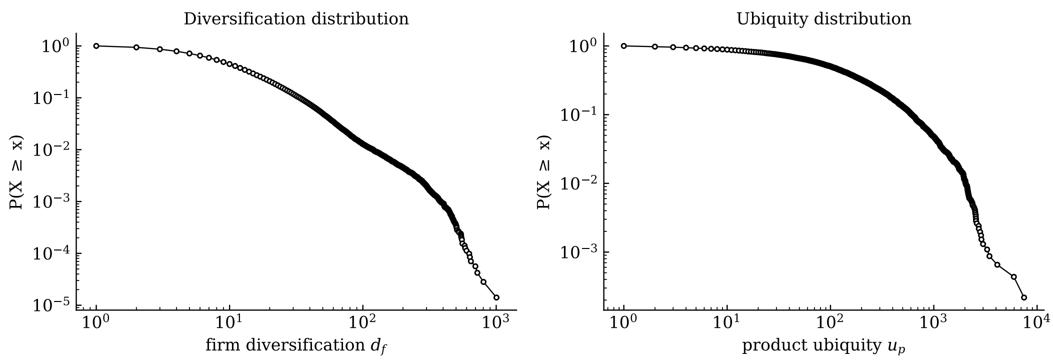


**Figure 5:** Complementary CDFs of firm diversification $d_f$ (left) and product ubiquity $u_p$ (right), log–log axes. Most firms are specialised and a few hugely diversified; most products niche and a few near-universal — the standard economic-complexity signature, which the block structure then organises.

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